Tunability of Sb_2Se_3 Phase Change Thin Films for Multi-Domain Optoelectronics

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stoichiometric = relating to or denoting quantities of reactants in simple integral ratios, as prescribed by an equation or formula.

Abstract

Phase change materials play an essential role in the development of tunable optics by overcoming the limitations possessed by conventional optics due to their static behaviour. Antimony-based chalcogenides (Group VIA) are promising due to low absorption loss, optimal refractive index change, and compatibility with silicon integration. This study optimises near-stoichiometric sputtered Sb_2Se_3 thin films and investigates their compositional, physical, and optical properties changes triggered by thermal, optical, and electrical stimuli. We further extend the investigation to the induced intermediate states in the Sb_2Se_3 thin film for multi-bit operation for tunable photonics and demonstrate phase-change memories with a current switching ratio of 10^4. The findings are integral to the realisation of high-capacity optical switches and routers, displays with larger colour distribution, and photonic memories with high computational capabilities.

Keywords

phase change materials, tunable properties, Sb_2Se_3, optoelectronics, multilevel switching, phase change memory

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Introduction

Dynamic photonic devices that are tunable and reconfigurable are a recent research focus because of the adaptiveness and smartness they can offer to potential applications. Common methods to introduce tunability of these devices include free carrier injection,[1] chemical and mechanical activation,[2] and using materials with large electro-optic,[3, 4] or optical nonlinear coefficients.[5, 6] However, these techniques suffer from slow speed,[5, 7] high power consumption, [8]requirements of long interaction lengths,[6] less temperature endurance, and minimal refractive index change[9] to achieve effective optical modulation.

Researchers have also been exploring phase-change materials (PCMs) to achieve a methodology for device tunability.[10-14] PCMs are materials that can transition between two structural states in response to externally applied stimuli. They possess unique properties, such as high optical contrast (Δn>1), low power consumption (nanojoules and lower), and fast switching speeds (nanosecond time scales). Thus, they can enable various applications, including optical computing, photonic memory, and optical neural networks.[15-19] Recently, extensive exploration in this area has mainly focused on $Ge_2Sb_2Te_5$ (GST)[20], $Ge_2Sb_2Se_4Te$ (GSST)[14], and vanadium dioxide (VO_2).[21-23] The phase transition in VO_2 is volatile and returns to its original state when it cools below the transition temperature. Hence, constant energy is required to retain the transition phase. The non-volatile nature and relatively significant refractive index change at 1550 nm (Δn~3) have made GST and GSST promising candidates for infrared (IR) applications for several years. Despite improvements in the material design, the absorption losses in crystalline and amorphous phases of these PCMs hinder their use in phase modulation schemes at the telecommunication band. Besides, it leads to the requirement of frequent amplification in the circuits and increases the complexity when designing cascaded devices.[24]

On the other hand, antimony-based chalcogenides PCMs, such as antimony trisulfide (Sb_2S_3) and antimony triselenide (Sb_2Se_3), exhibit almost zero absorption loss for both phases in the mid-IR region and have less loss over the visible wavelength compared to GST and GSST while maintaining an optimal refractive index change.[25-31] The transition between the amorphous and crystalline phases changes the arrangement of the chemical bonds in the material, resulting in a change in the optical properties, such as the complex index of refraction. In addition, the refractive index of this material system is close to that of silicon, which enables straightforward, direct integration with complementary metal oxide semiconductor (CMOS) systems.

Here, we report the optimisation of the Sb_2Se_3 thin films and the formation of intermediate phase states by different methods such as simple temperature annealing, nanosecond (ns), and femtosecond

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(fs) laser systems. We investigate the influence of deposition conditions on the obtained film stoichiometry and physical properties. We then comprehensively analyse the optical and electrical switching behaviour in the amorphous and crystalline phases of the thin film, including using intermediate states to realise multilevel switching. We could achieve both amorphisation and crystallisation using similar pulse width and demonstrate the material more compatible with industry needs. We also demonstrate Sb_2Se_3-based phase-change memory with an intermediate state, exhibiting repeatable switching between crystalline and amorphous states, achieving a high switching ratio of approximately 10^4. These findings contribute to a deeper understanding of the potential of Sb_2Se_3 for integrated photonics, particularly in the context of multi-bit operation.

Results and Discussion

Structural Properties and Thermal Switching of Sb_2Se_3 thin films

As outlined in the *Methods* section, a 60 nm thick Sb_2Se_3 thin film was deposited on a Silicon (Si) substrate using RF sputtering. A systematic study was conducted on the thin film to understand the influence of thermal annealing, and it was found that a fully crystalline phase was achieved after 8 min of annealing at 200 °C. The physical and chemical characterisation of the amorphous and crystalline phases of the thin film are illustrated in **Figure 1**.

Figure 1a depicts the Raman spectra of amorphous and crystalline Sb_2Se_3 thin film deposited on a Si substrate. The as-deposited material exhibits a broad peak at 190 cm^{-1}, typically attributed to the Sb-Se bond, indicating an amorphous structure. Following annealing/crystallisation, the Raman spectra showed multiple sharp peaks. In its crystallised form, Sb_2Se_3 thin film produces three A_g symmetry peaks at 117, 190, and 211 cm^{-1} and two B_{xg} peaks near 153 cm^{-1}.[32] Figure 1b displays the X-ray diffraction pattern of the crystalline Sb_2Se_3 (*c*-Sb_2Se_3) thin film. The sharp diffraction peak position obtained confirms the polycrystalline nature of Sb_2Se_3 with a preference for (020) orientation. All the peaks obtained for crystalline Sb_2Se_3 correspond to the orthorhombic phase.[33, 34]

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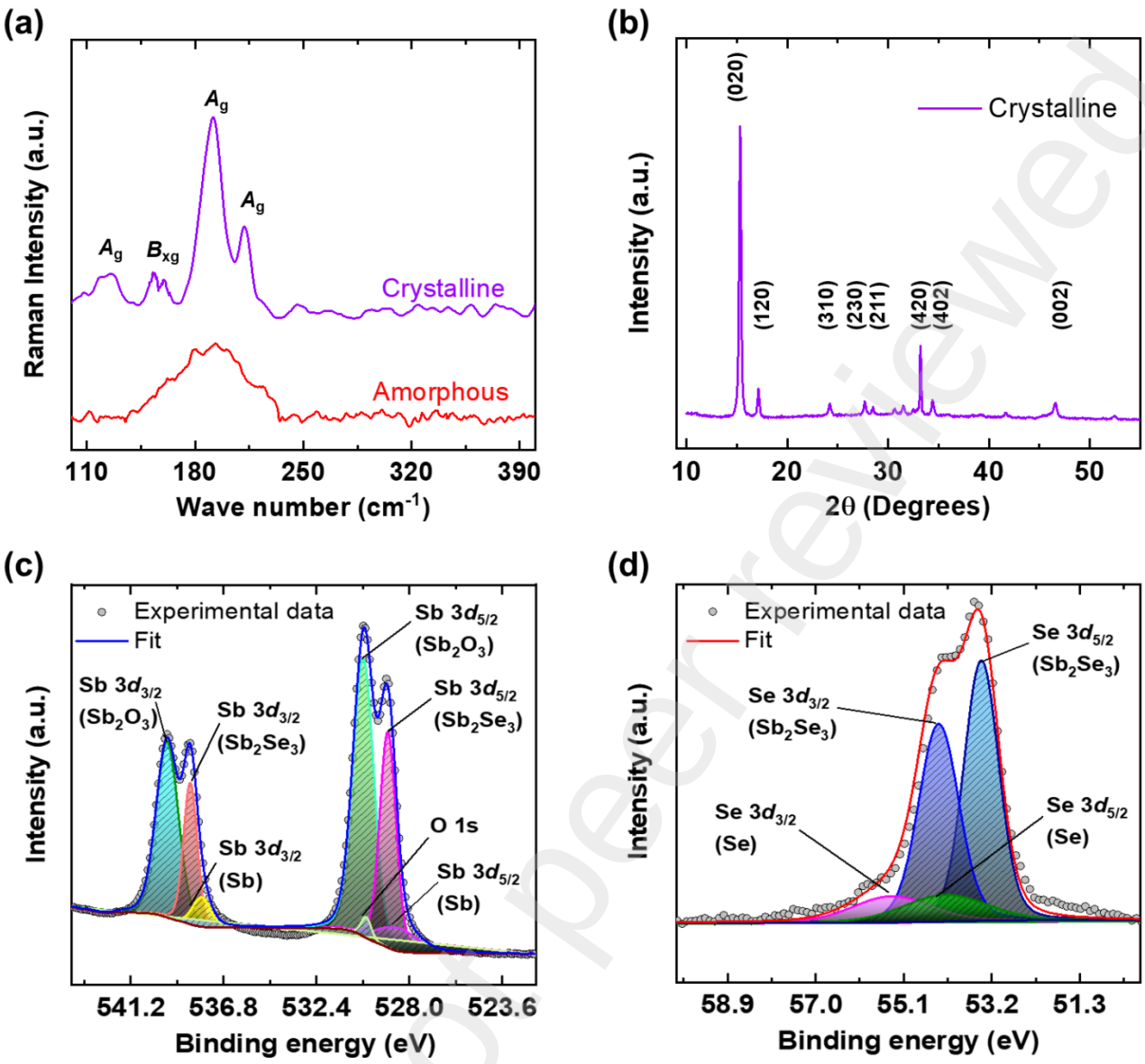

Figure 1 | Structural Characterisation. (a) Raman spectroscopy of amorphous (red) and crystalline (violet) Sb_2Se_3 obtained at 532 nm wavelength excitation. (b) X-ray diffraction pattern of *c*-Sb_2Se_3 thin film. Indexes of the diffraction peaks are marked for reference. (c, d) High-resolution XPS spectra showing core level spectra of Sb (3*d*) and Se (3*d*) in the crystalline film, respectively.

Stoichiometry for the amorphous and crystalline phases of the thin film was studied using XPS. The core level spectra of Sb 3*d* (523-544 eV) and Se 3*d* (50-60 eV) are shown in Figure 1c and Figure 1d, respectively.[35] The core level spectra of Sb $3d_{5/2}$ and Sb $3d_{3/2}$ in the crystalline film of Sb_2Se_3 are located at 529.84 and 539.22 eV, respectively, with a spin-orbit coupling of 9.38 eV; for Se $3d_{5/2}$ and Se $3d_{3/2}$ in the same film, the core level spectra are at 53.42 and 54.28 eV, respectively, with a spin-orbit coupling of 0.86 eV, which is quite consistent with reported values.[27] With this, the calculation of atomic percentage gives 43.61% for Sb and 56.39% for Se, with margin of error from the 2:3

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stoichiometry of Sb_2Se_3 (Sb = 40% and Se = 60%).[36] The effects of variable deposition power on the stoichiometry of the thin films were conducted using XPS, and the results are in the supplementary section. (Figure S1 and Table S1). It is observed that any deviations from the stoichiometry are often compensated by oxides of Sb, such as Sb_2O_3, which is evident from the high-resolution XPS spectra of Sb. Sb_2O_3 is more prominent for low power deposition which decreases with an increase in sputtering power. This is in line with Sb-rich conditions, where the quantity of Se is diminished, which allows for the unreacted Sb to get oxidised post-annealing under normal ambient conditions.[37] With an increase in sputtering power, the variations in stoichiometries were not as pronounced, which might indicate that being a single-phase compound, it does not permit dramatic scope in non-stoichiometric conditions. From this, for other characterisation studies performed, 70 W was chosen to be the optimal sputtering power, and all following experiments were conducted with the same.

The potential for investigating the properties of intermediate states in this material system to establish the multilevel/multi-bit operation[38] is essential for various applications in photonics and optical communication systems as this allows for parallel processing of information, results in several advantages such as increased capacity and assists complex data processing. Even though many studies have been conducted on materials such as GST, GSST, and Sb_2S_3 using either advanced laser systems with ultra-short pulses or integrated PIN (*p*-type, intrinsic, *n*-type junction) heater systems, there is not much evidence which suggests the potential for multilevel operation using Sb_2Se_3 thin film.[39, 40]

We focussed on using thermally regulated intermediate states in Sb_2Se_3 thin film. Creating multilevel states with thermal annealing reduces the complexity of photonic device design as it does not require additional circuitry either for laser coupling or electrical pulse excitation. To investigate the potential of forming multiple states, a 65 nm Sb_2Se_3 film was deposited on both Si and glass substrates. The crystallisation of the deposited film is done under varying annealing conditions. **Figure 2a** displays the transmission variation measured with a CRAIC microspectrophotometer, and the sample labels used in Figure 2 correspond to different annealing conditions of the film starting from the temperature 100 °C to 220 °C to achieve different levels of crystallinity. **Table 1** refers to the percentage of transmission change from *a*-Sb_2Se_3 at 633 nm and 1550 nm wavelength for each level of crystallinity. Detailed descriptions of the different annealing conditions are added in the Method section.

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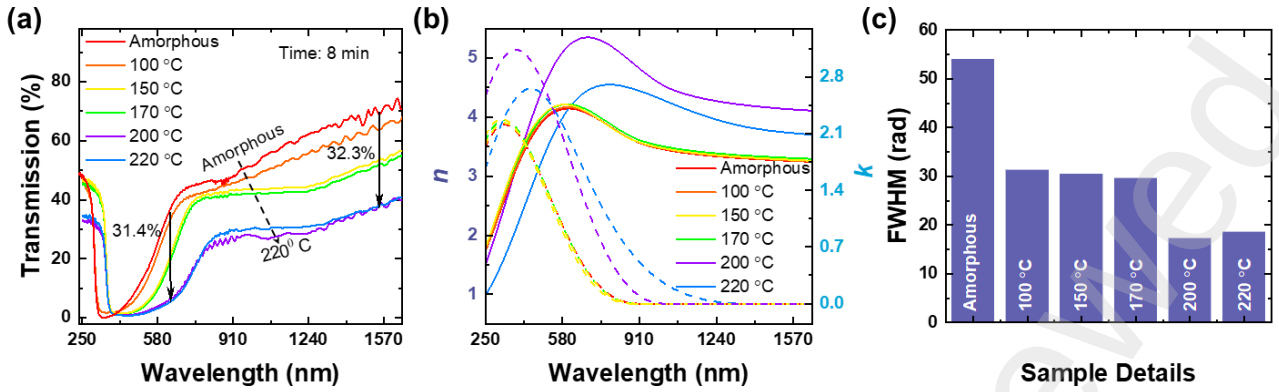

Figure 2 | Optical characterisation of Sb_2Se_3 thin film. (a) Transmission spectroscopy of Sb_2Se_3 films deposited on glass. The transmission changes 32.3% at 1550 nm when the film changes from amorphous to crystalline at 220 °C. The dashed black arrow indicates the red shift of the transmission minima with respect to the amorphous to crystalline transition. Annealing temperature corresponding to each intermediate state is indicated in the legend adjacent to the sample name within brackets. (b) The refractive index and extinction coefficient of Sb_2Se_3 films deposited on Si and annealed at different temperatures were measured in the 200-1600 nm wavelength range. (c) FWHM was calculated from the significant A_g peak at 190 cm^{-1} for each annealing condition to understand the degree of crystallinity of the films.

Table 1 | Transmittance Summary. Transmission changes in crystalline Sb_2Se_3 thin film from the amorphous sample at visible (λ = 633 nm) and infra-red (λ = 1150 nm) wavelengths.

	100 °C	**150 °C**	**170 °C**	**200 °C**	**220 °C**
λ = 633 nm	3.7	16.1	18.0	31.4	31.6
λ = 1550 nm	5.4	15.4	17.0	32.3	30.8

?

The transmittance is graphed across a wide wavelength range, from 250 to 1550 nm. A notable shift in the infrared transmittance can be seen as the Sb_2Se_3 film crystalline phase. In particular, the transmittance has ~32% change between *a*-Sb_2Se_3 and the samples annealed at 200 °C and 220 °C at 1550 nm, resulting in a total of 4 different levels of crystallinity. These samples also show four distinct states between amorphous to crystalline in the visible wavelength range, with a maximum reduction of 31.4% in the transmittance value at 633 nm as noted in Table 1. The observed change in the transmittance value with multiple levels suggests the possibility of realising a display technology that could represent a variety of colours for patterns/image generation, which could be possibly used for optical encryption. It is also important to notice in Figure 2a that there is a red shift in the transmission minima as the film is crystallised, indicating a decrease in energy levels associated with

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transitions between electronic states, implying the reduction in the optical bandgap during amorphous to crystalline transition.[41]

We then used an ellipsometer to measure the complex refractive index value of *a*-Sb_2Se_3 and *c*-Sb_2Se_3 across a wide spectral range of 200-1600 nm. Figure 2b displays the refractive indices, *n* (solid line) and extinction coefficients, *k* (dotted line) of samples with different annealing conditions as measured. The results were obtained by fitting the ellipsometer values using Cody–Lorentz oscillators to consider the absorption beneath the bandgap. Sb_2Se_3 thin films deposited on Si are employed for this analysis due to the ease of fitting and analysing. All the films show a *k* value of 0 at 1550 nm. This suggests that Δk is equal to zero when the film transitions from an amorphous to a crystalline phase, regardless of its degree of crystallisation. At 1550 nm, the sample annealed at 220 °C has a maximum change in the refractive index of 0.85, with a Δk of 0. The lack of loss in the Sb_2Se_3 thin film might be attributed to the decreased interband optical absorption.

When we examine the visible wavelength, specifically 633 nm, the refractive index difference (Δn) is 1.10 and 0.67 for samples annealed at 200 °C and 220 °C, respectively with 1.03 and 0.81 of the extinction coefficients difference (Δk), still very less compared to GST and GSST. Refractive index measurements have also revealed that when the temperature is raised above 200 °C while keeping the time constant, there is an increase in Δk value as in the 220 °C annealed sample. This could potentially be attributed to a more significant loss of selenium (Se) as the thermal budget is raised. Figure 2b clearly distinguishes at least four states due to different levels of crystallinity that could potentially be utilised in developing wavelength division multiplexing systems, optical interconnects, and reconfigurable optical add/drop multiplexers.

To understand the relation between the degree of crystallinity and transmission change, a full width at half maximum (FWHM) value was calculated from the obtained A_g Raman mode at 190 cm^{-1} for each sample. FWHM value decreases with increased crystallinity**.** Figure 2c exhibited a gradual decrease in the FWHM value as the annealing temperature and time increased, indicating the gradual crystallisation process in the Sb_2Se_3 thin film. It is worth noting from the diagram that samples annealed at 200 °C and 220 °C possess a similar FWHM value, significantly less than the *a*-Sb_2Se_3 thin film. The slight rise observed in the FWHM for the sample annealed at 220 °C aligns with earlier findings from complex refractive index measurements, suggesting that elevating temperature above 200 °C results in a decline in the crystalline quality of the film.

Switching mechanisms in Sb_2Se_3 thin film

Laser beam interactions with thin films can enable atomic-scale changes leading to high-resolution features and phase change within the system.[53] In recent studies, it was demonstrated that pulsed laser

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interactions can induce the phase change in non-volatile chalcogenide films like GST and GSST, for potential applications requiring high-resolution features like DVDs, logic circuits, and meta optics.[42]

As previously noted, Sb_2Se_3 is a material that can undergo reversible phase transformation between amorphous and crystalline states under certain conditions. This reversible phase transition can be activated by applying the appropriate external stimuli, usually laser heat or the heat induced by electrical pulses. Sb_2Se_3 can be crystallised by heating it above its glass transition temperature.[43] Generally, either hotplate heating or applying intensity and longer optical/electrical pulses are used for this purpose.

Conversely, amorphisation necessitates heating the material above its melting point and cooling the sample quicker than the crystallisation rate. This is typically accomplished by utilising high-intensity, shorter optical/electrical pulses.

Optically induced Sb_2Se_3 thin film using nanosecond laser beam interactions

We employed nanosecond (ns) laser beam pulses to investigate both crystallisation and amorphisation processes in Sb_2Se_3 thin film. For both processes, we maintained a comparable pulse width, but a different number of laser pulses per spot and different pulse energy. Crystallisation was achieved through multiple pulses at lower laser power, whereas amorphisation required single pulses at higher laser power.

The crystallisation and amorphisation are achieved by scanning a ns-laser beam with a full-width half-maximum of 50 µm and wavelength of 1064 nm. **Figure 3a** presents a schematic representation of the laser beam-induced phase change mechanism applied to Sb_2Se_3 thin films in this work. The depicted mechanism applies to both the amorphous (as deposited) and crystalline phases of the films, prepared under conditions identical to those used for samples annealed at 200 °C, on both silicon and glass substrates. As opposed to previously reported studies, we refrained from using a protective layer for this analysis[29, 30] to truly understand the changes due to the application of external stimuli, and it can be noted that the thin film survived the dose application with a resultant phase change behaviour.

To achieve the intermediate states and fully crystalline phase, we have varied the laser power from 3 W to 20 W with the pulse duration from 2 ns to 500 ns while varying the number of pulses per spot from 5 to 2250. We have found that the higher pulse width, 500 ns, and a larger number of pulses per spot, up to 2000, lead to the best result of crystallisation (Figure 3 and Table S2). Attempts to induce crystallisation with low pulse repetition rates and high energy pulses, even close to material ablation levels, yielded no observable results. This challenge may be attributed to the initial nucleation difficulties within the material system. Utilising multiple pulses served to condition the film initially, facilitating the crystallisation process once the initial nuclei were established.[44]

ablation = the removal or destruction of something from an object bty vaporization

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Figure 3b-d displays the AFM images of the film before and after crystallisation, which was achieved through thermal and laser-induced transformation. In a sample that has been thermally annealed, many crystal grains with various orientations can be seen due to the crystal growth being dominated by nucleation. Generally, for a laser beam treated samples demonstrate crystal growth in the direction of the laser spot movement during scans.[45]

A reversible study was conducted on the thermally crystallised Sb_2Se_3 thin film. We obtained the amorphous phase using an increased average laser power, *i.e.*, 20 W to 50 W and keeping pulse duration at comparable values to those for crystallisation, 350 ns (Supplementary Information Table S2). In the amorphisation study, all the laser parameters were adjusted to achieve only one pulse per spot. Applying multiple high-energy pulses for amorphisation will eventually result in the film ablation rather than a phase change mechanism.

In Figures 3e and 3f, we illustrate the change in reflectance during the processes of crystallisation and amorphisation of Sb_2Se_3 thin films induced by varying the power, pulse duration and number of pulses per spot by using nanosecond laser pulses. Samples which showed different degrees of crystallisation based on reflection measurements due to laser fluence are shown in Figure 3e. Laser fluence is defined as the applied laser energy per unit area and it is the function of laser power, frequency, and beam area. As it is observed in Figure 3e, increasing the applied laser fluence resulted in a gradual increase in the crystallisation of the film and at 0.713 J/cm^2 laser fluence a complete crystallization is achieved which is also corroborated by subsequent Raman measurements (Figure S3a). Applying laser fluence of more than 0.713 J/cm^2 resulted in damaging the film. It is also noted that lowering the laser fluence to 0.153 J/cm^2 did not result in any degree of crystallization as the corresponding temperature should be above 200 °C. In Figure 3e, the reflectance graph of 0.103 J/cm^2 laser fluence is the same as that of amorphous film. By controlling the laser parameter carefully, thereby varying the applied laser fluence to the film, we demonstrated six different states for Sb_2Se_3 thin films which have varying reflectance values.

Figure 3f displays the amorphisation of thermally crystallised Sb_2Se_3 thin films. Here, we have used higher laser fluence to achieve the re-amorphisation of the film as amorphisation requires heating the material above its melting point and then reducing the heat faster than the crystallisation speed. For Sb_2Se_3, the melting point is approximately 610 °C. As indicated in the figure, the lower laser fluence such as 3.18 J/cm^2 did not exhibit a phase change. As we increased the laser fluence to 8.49 J/cm^2 and above, it started showing the trend of amorphisation. We obtained optimal results for the laser fluence around 10 J/cm^2 and above. But increasing the laser fluence above 20 J/cm^2 resulted in the ablation of the thin film which was again confirmed by Raman spectra (see Supplementary S3b).

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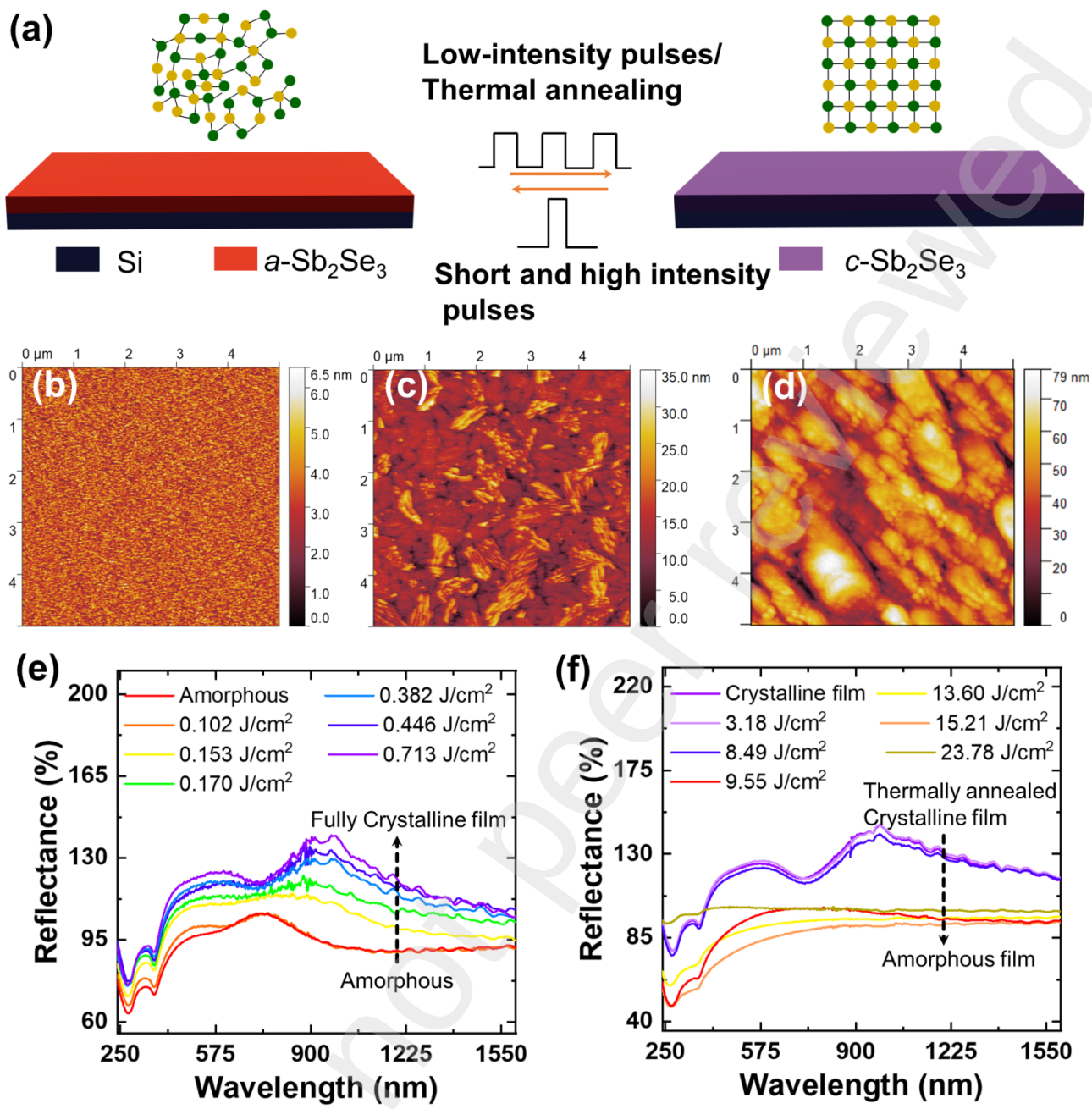

Figure 3 | Laser switching on Sb_2Se_3 thin film. (a) Schematic diagram explaining the mechanism behind the reversible phase change in Sb_2Se_3. AFM image showing the topology for (b) *a*-Sb_2Se_3 thin film, (c) *c*-Sb_2Se_3 thin film after thermal annealing, and (d) *c*-Sb_2Se_3 thin film after laser treatment. Reflectance spectra of (e) crystallisation of *a*-Sb_2Se_3, and (f) amorphisation of *c*-Sb_2Se_3 using nanosecond laser pulses. The arrow in the figures denotes the change in reflection during crystallisation and amorphisation. Reflectance % above 100 indicates that the crystalline Sb_2Se_3 thin film is more reflective than the reference taken in the measurement which is Si.

We successfully achieved both crystallisation and amorphisation of Sb_2Se_3 thin films using laser pulses with comparable pulse widths, 500 ns for crystallisation and 350 ns for amorphisation, and varying the laser fluence on the film. These results hold significant promise for utilising this material

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for tunability in photonic device applications and will allow fast creation of selectively crystallised areas on Sb_2Se_3 films.

Optically induced Sb_2Se_3 thin film phase change using femtosecond (fs) laser beam interactions

We also studied the influence of ultra-short fs-laser beams in the phase change behaviour of Sb_2Se_3 thin film using a visible wavelength, 532 nm, and a near-infrared (NIR) wavelength, 780 nm laser beam. Ultra-short pulses help control the photo-ionisation and thermal processes with the highest precision, inducing local photo modification in sub-100 nm-sized regions. However, there are recent approaches which can be used to tune the thermal effect using fs-laser beam pulses.[46, 47] The used 532 nm fs-laser beam has a pulse width of 50 fs, 70 MHz pulse repetition rate, and is labelled as 'fs-532 nm'. On the other hand, the 780 nm fs-laser beam has a pulse width of 100 fs, 80 MHz pulse repetition rate and is labelled as 'fs-780 nm'. All fs-laser beam interaction studies are carried out on Sb_2Se_3 thin film of thickness of 70 nm. The thin film undergoes rigorous treatment using different laser fluences to get the optimum phase change conditions and morphology. AFM imaging was conducted to understand the morphology changes in the fs-laser treated thin films, fs-532 nm and 780 nm and its reverse treatments are shown in Figure S4 and can be seen that there is a decreasing trend in morphology during the re-amorphisation process. Re-amorphisation of 'fs-532 nm' film is achieved by using 780 nm wavelength laser and indicated as 'fs-532 nm_rev' and that of 'fs-780 nm' is achieved by using 532 nm laser and labelled as 'fs-780 nm_rev'.

Figure 4a demonstrates phase change behaviour under laser beam treatment of both wavelengths and in reversible conditions. As mentioned, the as-deposited material exhibits a broad peak at 190 cm^{-1}, typically attributed to the Sb-Se bond, indicating an amorphous structure. After subjecting the fs-laser beam with a laser fluence of 0.63 J/cm^2 and 0.034 J/cm^2, Raman measurements revealed a phase change for both samples with a clear indication of A_g modes. However, the FWHM of 'fs-532 nm' thin film is the broadest, while that of 'fs-780 nm' is the sharpest indicating a stronger phase change. While, in reverse treated conditions, for the 'fs-532 nm_rev' thin film the peak becomes a boarder, which indicates a reversible phase change. By contrast, for the 'fs-780 nm_rev', the FWHM of the peak remains the same which indicates that no reversible phase change occurs. Notably, in contrast to thermally treated or ns-laser beam treated thin films, the reversibility was less pronounced in fs-laser beam treated thin films due to the insufficient heating of film resulting by too short pulses.

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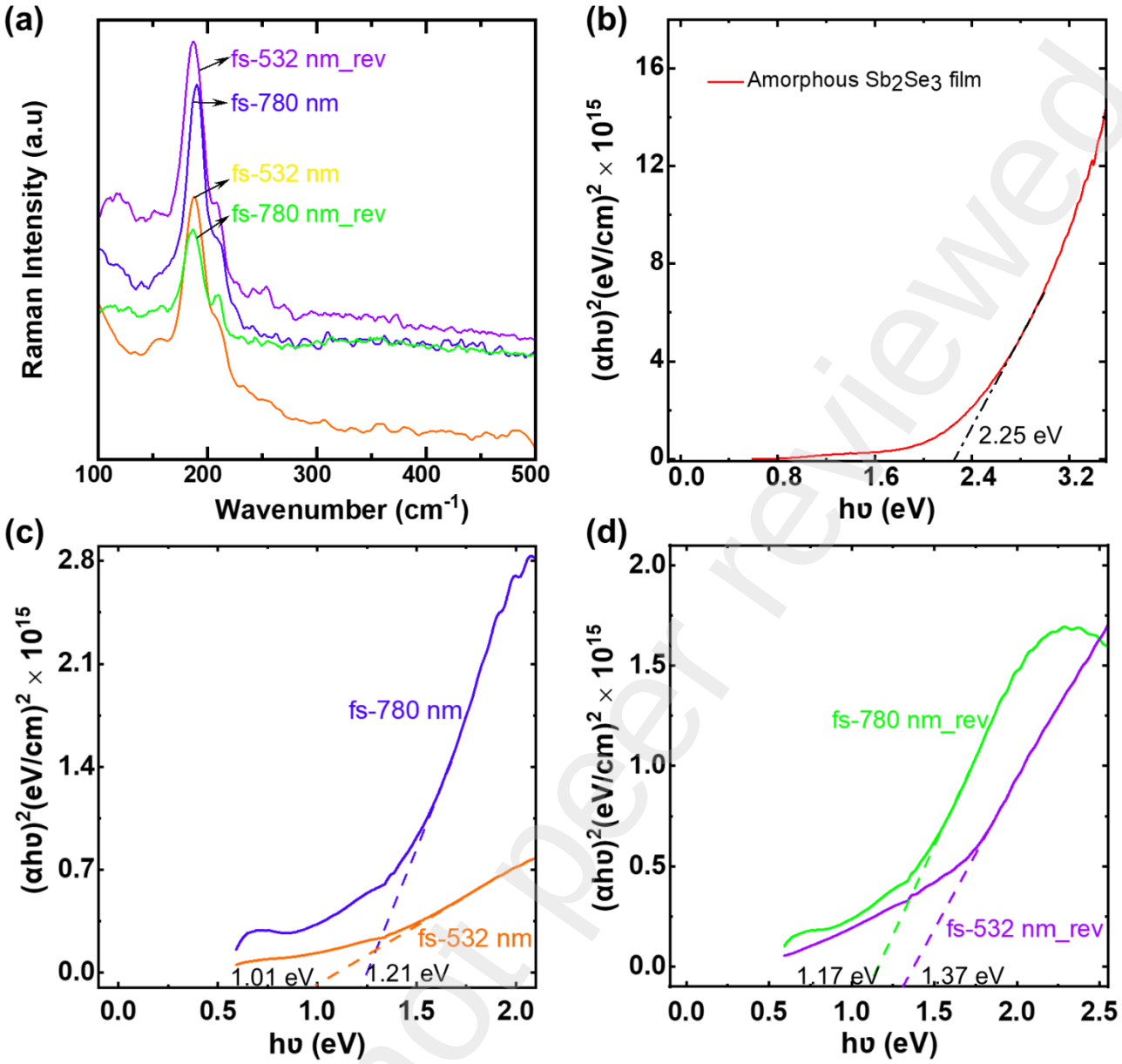

Figure 4 | fs-laser beam treated Sb_2Se_3 thin film characteristics. (a) Raman measurements on fs-laser treated and reverse treated thin film using wavelengths 532 nm and 780 nm. (b) Optical energy gap obtained for amorphous Sb_2Se_3 thin film. (c) Optical energy gap obtained for Sb_2Se_3 thin films, 'fs-532 nm' and 'fs-780 nm' after fs-laser beam treatment using 532 nm and 780 nm wavelengths. (d) Optical energy gap obtained for Sb_2Se_3 thin film, 'fs-532 nm_rev' and 'fs-780 nm_rev' after reverse fs-laser beam treatment.

Transmission measurements were conducted on 'fs-532 nm', 'fs-780 nm', 'fs-532 nm_rev', and 'fs-780 nm_rev' thin films (Figure S5a) and showed around 35% transmission change at the telecommunication wavelength of 1550 nm. Moreover, it was observed that the linear absorption coefficient is more significant after the laser treatment than the amorphous thin film. The observation can be due to different densities of localised defect states in the forbidden gap, which can affect the transitions of electrons. Further, we determined the optical bandgap for these thin films using Tauc's method as explained in the Supplementary Information. It was observed that the optical bandgap of

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the amorphous thin film is 2.25 eV, as given in Figure 4b. The calculated optical bandgaps of thin films 'fs-532 nm' and 'fs-780 nm' are 1.01 and 1.21 eV, respectively (Figure 4c). For the reverse wavelength treated thin films such as 'fs-532 nm_rev' and 'fs-780 nm_rev', the resultant optical bandgaps are 1.37 and 1.17 eV, respectively (Figure 4d). The obtained optical bandgaps for the thin film under different conditions confirm that the trend of phase change from amorphous to crystalline and *vice versa* are more prominent with 780 nm wavelength laser where the more thermal effect is pronounced.

The impact of thermal energy is more significant in the near-infrared (NIR) and longer wavelength regions compared to visible wavelengths.[46] In the specific scenario mentioned, where 780 nm and 532 nm wavelengths are employed, the use of a 780 nm femtosecond laser beam on Sb_2Se_3 thin film previously treated with a 532 nm wavelength laser suggests that the phase change in the thin film is primarily influenced by thermal effects. This is evidenced by the observed reversibility in the phase change process. This is further evidence of phase change from crystalline to amorphous nature with the practical tuning of laser irradiation conditions, which will have high potential in realising miniaturised tunable devices.

Optical nonlinear properties of Sb_2Se_3 thin film using femtosecond (fs) laser beam interactions

Further, we studied the nonlinear optical (NLO) properties of the Sb_2Se_3 thin film of thickness, 70 nm, using fs laser beam interactions with the wavelength of 790 nm with a pulse width of 140 fs, and 80 MHz pulse repetition rate. We chose the mentioned laser beam irradiation parameters based on the optimum conditions obtained for the fs-laser beam interactions with the Sb_2Se_3 thin film to obtain a phase change in the thin film. A modified open aperture (OA) *z*-scan method was employed to conduct the NLO studies, and a schematic of the measurement setup was given in Figure S6a, and measurement details were given in the *Methods* section.[50] The recorded CMOS image of the transmitted beam through the thin film positioned at 20 mm from the focal point is given in **Figure 5**a and 5b for amorphous and crystalline thin film, respectively. The thin films were sequentially moved along the *z*-axis for the OA measurements to avoid nonlinear scattering. The transmitted intensity for OA was normalised with the far-field intensity to obtain normalised transmittance induced by laser fluences from 0.034 J/cm^2 to 0.086 J/cm^2. The experimental data was fitted to the equation below for both amorphous and crystalline thin films.[51]

$$T = \frac{1}{\sqrt{\pi\left[\frac{\beta I_0 L_{eff}}{\left(1+\frac{z^2}{z_0^2}\right)}\right]}}\int_{-\infty}^{+\infty} \ln\left[1+\frac{\beta I_0 L_{eff}}{1+\frac{z^2}{z_0^2}} exp(-t^2)\right]dt \quad (3)$$

where T is the normalised transmittance, I_0 is the peak on-axis irradiance at the focus, z_0 is the Rayleigh length, and L_{eff} is the path length of the sample. It was observed that the normalised

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transmittance decreases when the position of the thin film approaches the focal point, leading to a symmetric valley in the *z*-scan curve for the amorphous thin film, as shown in Figure S6b which indicates the reverse saturable absorption (RSA) response. In contrast, the transmittance increases, indicating a symmetric peak in the crystalline thin film, indicating a saturable absorption (SA), as shown in Figure S6c. It was also observed that both amorphous and crystalline thin films showed large modulation depths under exposure to higher laser fluences.

Further, the nonlinear absorption, β_{eff}, a concentration-dependent parameter, was calculated with the data fitting using the formula as mentioned in *eq*.3 for the thin film with both amorphous and crystalline conditions using 0.034 and 0.086 J/cm^2 laser beam fluences. In the Z-scan experiment, if the material shows a $\beta_{\mathrm{eff}} > 0$, the behaviour corresponds to RSA and $\beta_{\mathrm{eff}} < 0$ responds to a SA behaviour in the thin film.[51] It was observed that the β_{eff} increases in the crystalline, obtained after annealing with a change of sign from positive to negative, representing a NOA behaviour change from RSA to SA. The 830 and -1800 cm/GW obtained β_{eff} values for amorphous and crystalline thin films indicate a switching behaviour from RSA to SA, which can be compared to the reported in the recent literature.[49, 50] It was observed that the β_{eff} increases in the crystalline, obtained after annealing with a change of sign from positive to negative, representing a NOA behaviour change from RSA to SA.

Besides, the optical limiting (OL) properties of these two films were studied from the OA-*z*-scan measurements using the position-dependent input laser beam fluence. The OL threshold can be defined as the input laser beam fluence for which the sample transmittance is 50%. The input fluence can be calculated from the slope of the output intensities (I_{out}) plotted as a function of the input intensities (I_{in}) given in Figures 5c and 5d for the amorphous and crystalline thin film. It can be seen in Figure 5c that the slope of the amorphous thin film decreases, which indicates an optical-limiting behaviour. On the contrary, as shown in Figure 5d, the slope of the I_{out}–I_{in} graph for crystalline thin film can be observed to be gradually increasing, which can point to the increase in the transmittance. In addition, the linear behaviour for the I_{out} at the lower laser fluences for both thin films slowly deviate as the I_{in} increases. This indicates the exhibition of RSA and SA behaviour at higher input laser fluences for the amorphous and crystalline thin films. The distinctive RSA and SA responses demonstrate the potential of Sb_2Se_3 thin film for use in different device-oriented applications like optical-power limiters and passive mode-locking devices.[48]

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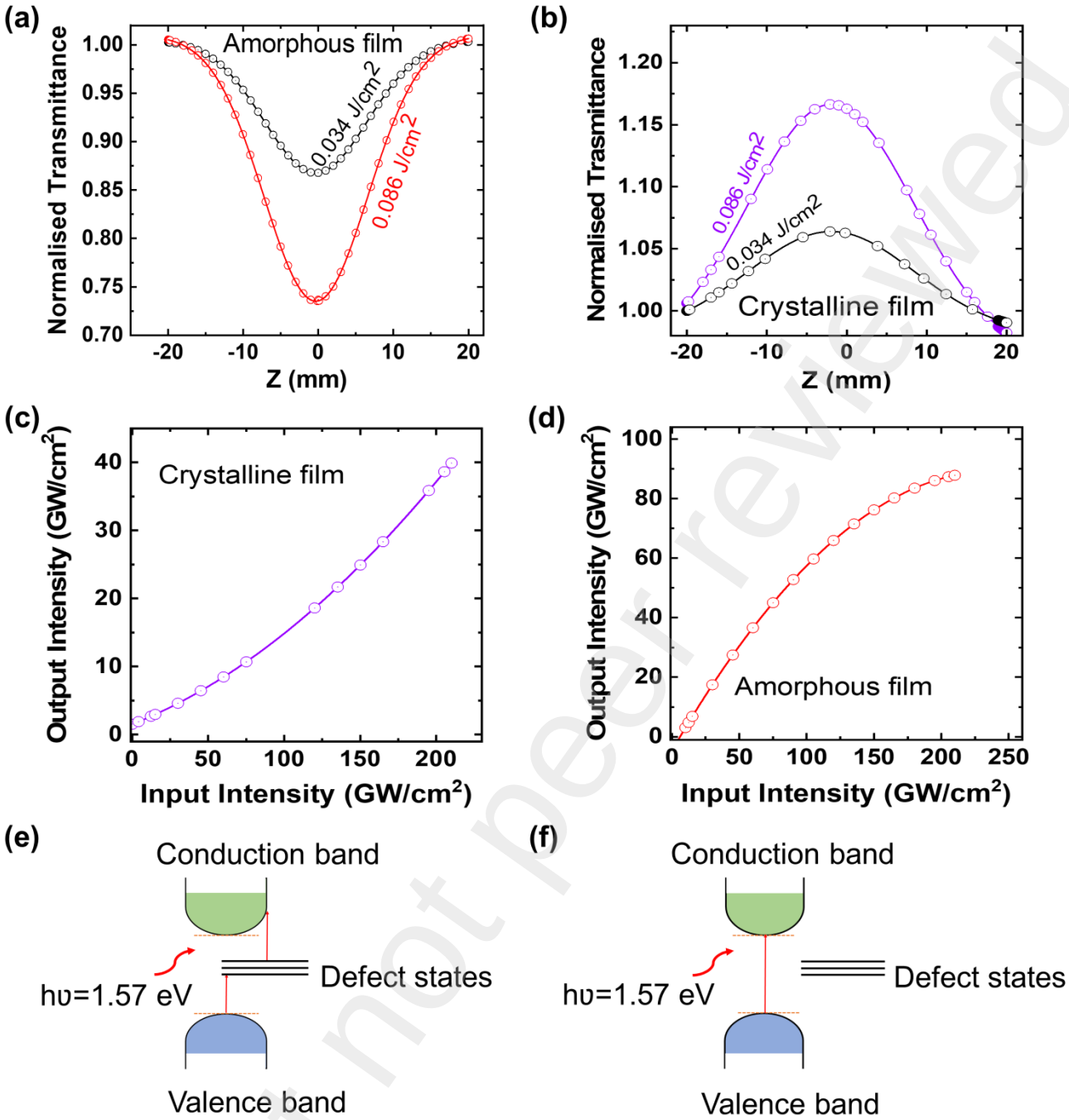

Figure 5| Nonlinear characteristics of Sb_2Se_3 thin film. (a) Normalised transmittance curve of amorphous thin film for open aperture *z*-scan. (b) Normalised transmittance curve of crystalline thin film for open aperture *z*-scan. (c) I_{out}–I_{in} curve for amorphous thin film. (d) I_{out}–I_{in} curve for crystalline thin film. (e) The proposed band diagram for the 3PA process in amorphous thin film. (f) The proposed band energy diagram for crystalline thin film.

Based on the experimental results, we can demonstrate three-photon absorption (3PA) during the fs-laser beam induced phase change in the Sb_2Se_3 thin films, probably induced by the electronic transition of the tunable behaviour as the result of localised defect states in the thin film due to unsaturated bonds of the material. These defect states are responsible for the electron transition from the valence band to the conduction band. The presence of defect states is more in the amorphous thin

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film than in the crystalline thin film, which can be concluded from the observation of the optical band gap of the amorphous Sb_2Se_3 thin film compared to that of the crystalline Sb_2Se_3 thin film. In this condition, electrons can transfer from the valence band to defect states by absorbing excited photons. However, at the same time, some electrons in the defected states can be transferred to higher excited states in the conduction band by absorbing other photons. Whereas in crystalline Sb_2Se_3 thin film, the electrons can directly transfer from the valence band to the conduction band by absorbing excited photons, which results in the conduction band almost occupied condition, and the Pauli-blocking process avoids further absorption of photons leading to SA response by the thin film.[51] We have demonstrated a possible defect state-induced 3PA process in the thin film for amorphous thin film in Figures 5e and 5f for amorphous and crystalline Sb_2Se_3 thin film.

Electrical switching of Sb_2Se_3

As a proof of concept, we have also demonstrated the electrical switching of Sb_2Se_3 thin films. To realise multiple states in Sb_2Se_3 thin film electrically, we have used two methods. They are (i) incorporating an indium tin oxide (ITO) heater to the Sb_2Se_3 thin film and (ii) realising a phase change memory using Sb_2Se_3 thin film. As schematically shown in **Figure 6**a, we have used the bottom ITO heater approach and deposited alumina (Al_2O_3) as the protective layer to avoid Se evaporation during the heating process. Gold (Au) contacts are patterned to apply electrical pulses to the ITO heater. Supplementary Figure S7a represents the optical image of the as-fabricated device. The fabrication process and thickness details are mentioned in the *Methods* section. Figure S7b in supplementary information represents the Raman spectra obtained from the heater when the film is in amorphous and crystalline states.

As these devices were fabricated on silicon dioxide (SiO_2) coated Si substrate, we measured reflectance spectra and illustrated in Figure 6b to confirm the phase change. The change in the reflectance is focused on 300-500 nm wavelength as the reflectance from Au/ITO/Al_2O_3 is very high at higher wavelengths. The background reflectance spectra from other layers in the structure are added in the supplementary (see Figure S7c). As expected, there is a change in the reflectance with the difference in its crystallinity according to the energy of the applied pulse. The legend used in Figure 6b follows the 'energy per pulse (number of pulses applied)' notation. As indicated in Figure 6b, the application of a single pulse with energy 8.5 mJ resulted in the full crystallization of the Sb_2Se_3 thin film.

Phase change memory using Sb_2Se_3 thin film

The ability of PCMs to switch and tune has drawn considerable attention due to their potential use in integrated photonics and electronics. In this context, phase-change memory can be used as a storage

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component for photonic data in integrated photonic circuits.[17, 52-55] Phase change memory is a form of non-volatile memory technology that uses the reversible phase transition of a material to store and access data.[56] This is due to phase change memory's capacity to alter physical characteristics, such as electrical conductivity, optical reflectivity, or optical transmission when transitioning between different phases.

The operational principle underlying phase change memory is illustrated in supplementary information in Figure S8a. Generally, phase-change memory devices comprise a thin layer of PCM sandwiched between two electrodes. By applying electrical pulses, the state of the material can be changed from amorphous to crystalline. The contrasting electrical resistance between the highly conductive crystalline phase and the low-conductive amorphous phase of the phase change material allows for data storage (WRITE operation). The information stored in phase change memory can be accessed by measuring the device's electrical resistance (READ operation). The WRITE process of phase change memory is heavily reliant on the temperature variation across the phase-change material after the voltage pulse is applied. Studies have demonstrated that the highest temperature is very near the bottom electrode in a typical device.[52] As a result, a mushroom-like structure is formed during amorphisation. The amount of READ resistance/current measured will depend on the size of the mushroom structure.

A schematic of the fabricated device is shown in Figure 6c, and the optical image is added in the supplementary section (Figure S8b). As illustrated in the schematic, 90 nm Sb_2Se_3 thin film was deposited between the titanium/platinum (Ti/Pt) bottom electrode and the ITO top electrode. ITO was used both as the top electrode and as the protective layer for the Sb_2Se_3 thin film. The fabrication process is outlined in the *Methods* section. The electrical measurements were conducted using a Keithley 4200SC. Figure 6d illustrates the typical current-voltage (I–V) characteristics of the amorphous and crystalline states of the Sb_2Se_3 thin film. The crystalline phase shows ohmic behaviour at low voltages, while the amorphous state displays nonlinear characteristics in the relation between current and applied voltages. After the threshold voltage (V_{th}) reaches 3.35 V, the conductivity in the amorphous phase increases quickly and exhibits a slight differential negative resistance mechanism, also known as voltage snapback (blue circle in Figure 6d).[57]

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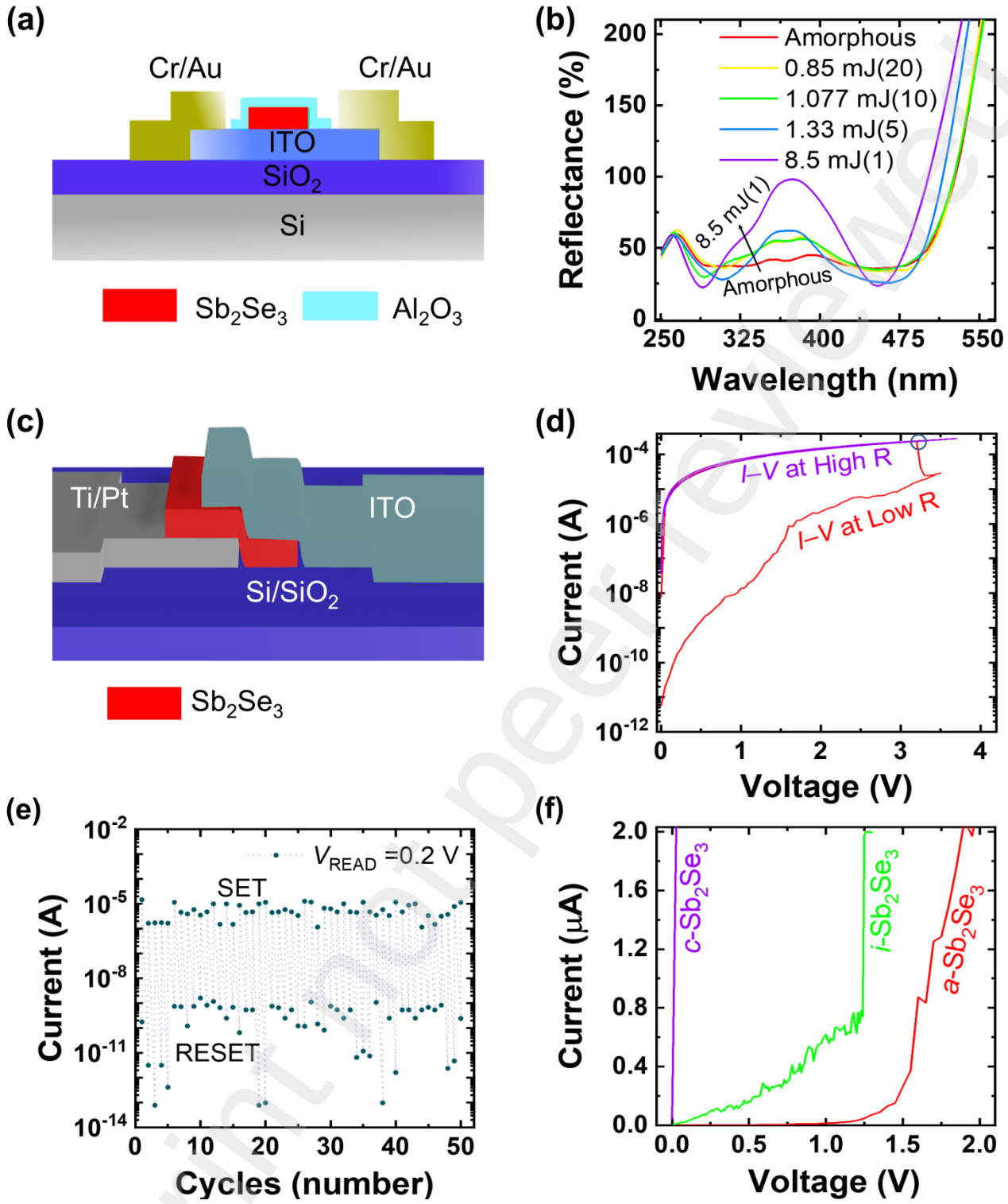

Figure 6 | Phase change memory device. (a) Schematic diagram representing the cross-sectional view of the heater incorporated device. (b) Measured reflectance before and after electrical pulses on the device junction area. The violet line shows almost two times increase in the reflectance for 370 nm wavelength. The legend used shows the 'energy per pulse (number of pulses applied)' (c) Schematic of the fabricated phase change memory device in this work. (d) The red line represents the *I–V* characteristics of the Sb_2Se_3-phase change memory device initially in an amorphous state. The violet line shows the *I–V* characterisation starting from the crystalline state. The proper merge between the red and violet lines confirms the total crystallisation of the

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device. (e) Change in current in Sb_2Se_3-phase change memory device during SET and RESET operations. (f) Measured *I–V* characteristics after RESET state for three different resistance states.

The state of the film just after threshold switching is usually called the amorphous ON state, implying that the amorphous phase has not yet crystallised. After allowing an adequate amount of current to flow through the device in the amorphous ON state, a memory switch occurs, leading to complete crystallisation. This causes the current-voltage characteristics of the amorphous ON (red) state to become indistinguishable from those of the crystalline state (violet). Figure 6e displays the current data obtained from the subthreshold region during the SET and RESET cycles. To amorphise the Sb_2Se_3 thin film, a 9 V, 200 ns electrical pulse with a 40 ns rise and fall time is applied after the crystallisation process. As mentioned earlier, higher voltage and the sharp edge of the pulse are required to raise the temperature above melting point as well as to ensure the sudden cooling to prevent recrystallization of the film. To transition from the amorphous to the crystalline state (SET) in Sb_2Se_3 thin film, a voltage pulse of 5 V and a pulse width of 600 µs is applied to raise the temperature within the PCM device to a level where rapid crystallisation takes place. A longer pulse width is necessary for complete crystallisation in the contact area. This process results in a high-conductance or SET state, which is characterised by a crystalline structure. We selected 0.2 V as the read voltage (V_{READ}) to prevent undesired crystallisation during the READ operation. The supplementary information (Figure S8d) includes an *I–V* sweep from 0 V to 0.2 V. We observed a four-fold decrease in the current between the SET and RESET operations over 50 cycles, which provided a clear distinction between the two states (Figure 6e). Phase change memory devices are renowned for their excellent retention. This is the capacity of the memory cell to maintain the stored data over a period. This also applies to the stability of the crystalline and amorphous forms of Sb_2Se_3. Figure S8c in supplementary information illustrates the retention test on the Sb_2Se_3 device, which demonstrated stable phases and no variation of switching ratios for more than 5000 s (>83 min).

Furthermore, the tests conducted on multiple devices with different bottom electrode widths indicate that the disparity between the bottom and top electrodes is essential in creating a phase-change memory device with a high switching ratio. When the gap between the bottom and top electrode dimensions of a device is small, the switching ratio is observed to be relatively low. Figure S9 in supplementary information includes the subthreshold regime *I–V* characteristics of one such device. By employing a smaller bottom electrode, a higher current density can be attained in PCM devices, resulting in more excellent electric power dissipation within the phase change material close to the bottom electrode. This localised energy dissipation leads to a rise in temperature close to the bottom electrode region. If we denote T_{hs} as the temperature hotspot in the region just above the bottom electrode, we can express it as:

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$$T_{hs} = R_{th}P_{in} + T_{amb} \quad (4)$$

where R_{th} is the average thermal resistance of the transported heat. P_{in} and T_{amb} are the input power and the ambient temperature, respectively.

We also attained an intermediate state between the amorphous and crystalline states by adjusting the characteristics of the SET electrical pulse. Here, we used a 4 V rectangular pulse with a duration of 200 µs. Figure 6f illustrates the *I–V* characteristics of the three distinct resistance states (referred to as *a*-Sb_2Se_3, *i*-Sb_2Se_3, and *c*- Sb_2Se_3) and clearly shows the variation in the threshold voltage to the crystallinity of the Sb_2Se_3 thin film, indicating that it can be used for multilevel operations. The results obtained from the characterisation open up the possibility of using Sb_2Se_3 as a promising material for memory devices with simple fabrication processes which are compatible with CMOS technology.

Table 2 denotes the summary of all external stimuli used in this paper, *i.e.*, thermal annealing, ns and fs laser systems, electrical heater, and phase change memory to achieve the repeatable crystalline, intermediate, and re-amorphise states of the Sb_2Se_3 thin film.

Table 2 | Summary of different stimuli on Sb_2Se_3 phase change. Summary of the different stimuli conditions used in this paper to achieve intermediate, crystalline, and re-amorphise state of the Sb_2Se_3 thin film.

Stimuli	***a*-Sb_2Se_3**	***i*-Sb_2Se_3**	***c*-Sb_2Se_3**
Thermal Annealing	• Room temperature deposition • No re-amorphisation	• Annealing between 100 °C to 200 °C for 8 min	• Annealing at 200 °C for 8 min
ns-laser	• Room temperature deposition • Re-amorphisation: laser pulses with energy fluence of 9.55 J/cm^2	• Laser pulses with energy fluence of 0.102 J/cm^2 to 0.446 J/cm^2	• Laser pulses with an energy fluence of 0.713 J/cm^2
fs-laser	• Room temperature deposition • Could not achieve re-amorphisation	• Intermediate states are not realised.	• 0.63 J/cm^2 with 532 nm laser and 0.034 J/cm^2 with 780 nm laser
Phase change memory	• Room temperature deposition. • Re-amorphisation: 9 V, 200 ns electrical pulse	• 4 V, 200 µs electrical pulse	• 5 V, 600 µs electrical pulse

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Finally, comparing the energy consumption value of all the external stimuli we have applied such as simple thermal annealing, ns and fs laser beam irradiation, and electrical heater, the lowest energy consumption is with fs laser treatment with 780 nm wavelength, and this also can meet the resolution requirements from industry. However, a complete reverse transition is not achieved with the fs-laser beam irradiation. Whereas the proper control of the ns laser system can trigger crystallisation and re-amorphisation of the Sb_2Se_3 thin film with a favourable amount of energy consumption meeting industry requirements.

Conclusion

In summary, we have successfully optimised Sb_2Se_3 thin films, a near-stoichiometric zero-loss phase change material, specifically tailored for a 1550 nm telecommunication wavelength. These films exhibit responsiveness to thermal, optical, and electrical stimuli, allowing for transitions between multiple states. Notably, the development of a phase change memory cell using Sb_2Se_3 thin films demonstrated a significant four-fold change in conductivity during SET and RESET operations. With consistently adjustable characteristics and low absorption loss, these films offer promising opportunities in optical imaging, optical switching, optical sensing, and optical communications. Additionally, their suitability for phase change memory cells, coupled with a multilevel operation system and zero static power, holds potential for applications in optical computing and quantum information processing. Overall, our research underscores the versatile nature of Sb_2Se_3 thin films, opening avenues for advancements in various technological domains.

Materials and Methods

Deposition of Sb_2Se_3 thin film

We have used a 99.99% pure Sb_2Se_3 target from Testbourne Ltd. Thin films of 60 nm Sb_2Se_3 thin film deposition were optimised using varying RF power from 50-90 W under argon (Ar) atmosphere. All the experiments are conducted on the films deposited at 70 W power using an RF source of *Kurt J Lesker* sputterer for 12 min to get a 60-65 nm film deposition which is confirmed by Bruker-DektakXT-Stylus profilometer.

Material characterisation

Atomic force microscopy (AFM) in tapping mode was performed on the film to identify the morphology of the amorphous and crystalline Sb_2Se_3 thin film (*Bruker Dimension Icon*). Raman

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measurements were performed on the films/devices using a 532 nm laser at a power of 5 mW, 50× objective, and with 1800 gr/m gratings (*Horiba LabRAM HR Evolution*). Further, Transmission/Reflection spectroscopy was performed on different locations of the film and devices with incident light of 20 µm spot size (*CRAIC Apollo Microspectrophotometer*). To confirm the crystallization phase of the Sb_2Se_3 thin film, XRD was carried out using an X-ray diffraction powder analyzer-D2 phase, Bruker. The obtained results were matched with ICDD No: 015-0861 confirming the orthorhombic phase formation in the film.

X-ray photoelectron spectroscopy data was collected using a *Thermo Scientific K-Alpha* X-ray photoelectron spectrometer. A monochromatic Al Kα X-ray source with an excitation energy of 1486.7 eV was used as the radiation source with a spot size of 400 µm (in 5 µm steps). A flood gun was used to neutralise the surface before starting the experiment to prevent any charge energy shifting. The core-level, survey and valence band spectra were collected under a high vacuum. Adventitious carbon affixed to 284.8 eV was used for peak correction. Peak fitting and data analysis were carried out using *CasaXPS* software. To prevent any energy shift, a flood gun was used to neutralize the surface before the experiment. All subsequent scans were performed under a high vacuum, with adventitious carbon (C 1*s*) attached to 284.8 eV for peak correction.

Optical switching characterisation

To study the control of Sb_2Se_3 thin film by ns laser pulses, we employed a 1064 nm laser system (MOPA7 Mactron) with adjustable pulse widths ranging from 2 ns to 500 ns and pulse repetition rates spanning from 20 kHz to 2000 kHz. To manipulate the number of pulses applied to specific spots on the samples, we used a computer-controlled Galvo scanner (Sino-Galvo) with adjustable scanning speeds. We selected a scanning speed range of 10 to 7000 mm/s, enabling precise control over the number of pulses per spot, varying from 1 to 2250.

For conducting fs-laser beam interaction with Sb_2Se_3 thin film, we have used 532 nm with a pulse width of 50 fs, 70 MHz pulse repetition rate (*Coherent Fidelity*) and 780 nm wavelength with a pulse width of 100 fs, 80 MHz pulse repetition rate (*Nanoscribe*) for fs-laser beam interaction studies to understand the linear optical properties.

To measure the nonlinear optical absorption of Sb_2Se_3 thin film, we employed a single beam open-aperture (OA) mode Z-scan setup using an fs-laser beam having 140 fs pulse width, 70 MHz repetition rates (Coherent Chameleon laser systems) using a wavelength of 790 nm. Using a stepping

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motor (Thorlabs), the samples were moved along the propagation direction (z-axis). The intensity distribution of the transmitted beam was recorded in a CMOS camera (Andor Marana CMOS).

Electrical characterisation

For the optical property tuning of Sb_2Se_3 thin film with electrical stimuli, we have fabricated a bottom ITO Heater with a thickness of 50 nm and a dimension of 60×100 μm^2. ITO deposition was conducted at 90 W power using an RF sputterer under an Ar atmosphere. A 60 nm thick Al_2O_3 thin film was deposited using the electron beam evaporation technique as the protective layer. The deposition of Al_2O_3 thin film was to prevent the Se evaporation from the film surface during the phase change process. All the electrical pulses are applied using a Keithley 4200SCS unit.

Phase change memory fabrication

Devices were fabricated on a pre-cleaned SiO_2/Si wafer using a two-step process involving photolithography and lift-off techniques. In the first step, the metal electrode is defined using AZ5214E photoresist using Heidleberg instruments *Maskless Aligner 150*. Subsequently, a 30 nm layer of Pt with a 5 nm Ti adhesion layer was deposited using an electron beam evaporator system (Kurt J. Lesker PVD 75), followed by a lift-off process to create the electrode pattern. Pt is ideal for the bottom electrode due to its high-temperature stability. In the second step, phase change material was patterned similarly, and a 90 nm Sb_2Se_3 film was sputtered using the RF source of Kurt J. Lesker sputter machine under an Ar vacuum. 90 nm thick Sb_2Se_3 films are used to avoid any leakage current through the grain boundaries in memory devices. Finally, the top electrode was patterned by a photolithography process, followed by a deposition of 130 nm thick ITO film to enable ease of optical characterisation. All the electrical measurements for the phase change memory are conducted using the Keithley 4200SCS unit.

Supporting Information

Additional characterisation results, experimental details, calculations, and measurements (PDF)

Acknowledgements

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Data Availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

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